

PROJECT TITLE

Crime Analytics Dashboard (USA - 2020)

1. Project Overview and Objective

1.1 Project Overview

This project involves cleaning, transforming, and analyzing crime data reported across the USA during **2020** using **Microsoft Excel** and developing an interactive dashboard in **Microsoft Power BI**. The dashboard provides insights into crime trends, victim demographics, weapon usage, case status, and crime distribution to support data-driven decision-making.

1.2 Project Objective

The main objective of this project is to demonstrate data preprocessing using Excel and create an interactive Power BI dashboard to analyze crime data and generate meaningful business insights for better decision-making.

2. Data Sources

- **Dataset Name:** Crime Data
 - **Dataset Source:** Public Safety / Law Enforcement Dataset
 - **Data Timeline:** 2020
 - **File Format:** CSV
 - **Analysis Tools:** Microsoft Excel & Microsoft Power BI
 - **Domain:** Crime Analytics | Public Safety | Business Intelligence
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3. Project Objectives

- Analyze crime trends across different reporting areas.
- Identify the most common crime categories.
- Study victim age and gender distribution.
- Analyze monthly crime patterns.
- Monitor open and closed investigation cases.
- Analyze weapon usage in crime incidents.
- Develop an interactive Power BI dashboard.
- Generate descriptive, diagnostic, predictive, and prescriptive insights

4. Attribute (Column / Feature) Details

Attribute Name	Data Type	Description
Division of Records Number	Integer/Text	Unique crime record ID
Date Reported	Date	Date when crime was reported
Date Occurred	Date	Date when crime occurred
Time Occurred	Time	Time of crime occurrence
Area Name	Text	Reporting area name
Crime Code	Whole Number	Crime identification code
Crime Description	Text	Description of crime
Victim Age	Whole Number	Age of the victim
Victim Gender	Text	Gender of the victim
Weapon Description	Text	Weapon used in the crime
Case Status	Text	Investigation status
Crime Location	Text	Crime occurrence location

5. Tools & Technologies

- **Microsoft Excel**
 - Data cleaning
 - Data transformation
 - Data validation
 - Data type conversion
 - Dataset preparation
- **Microsoft Power BI**
 - Data modelling
 - Relationship creation
 - DAX calculations
 - Dashboard development
 - Interactive visualizations

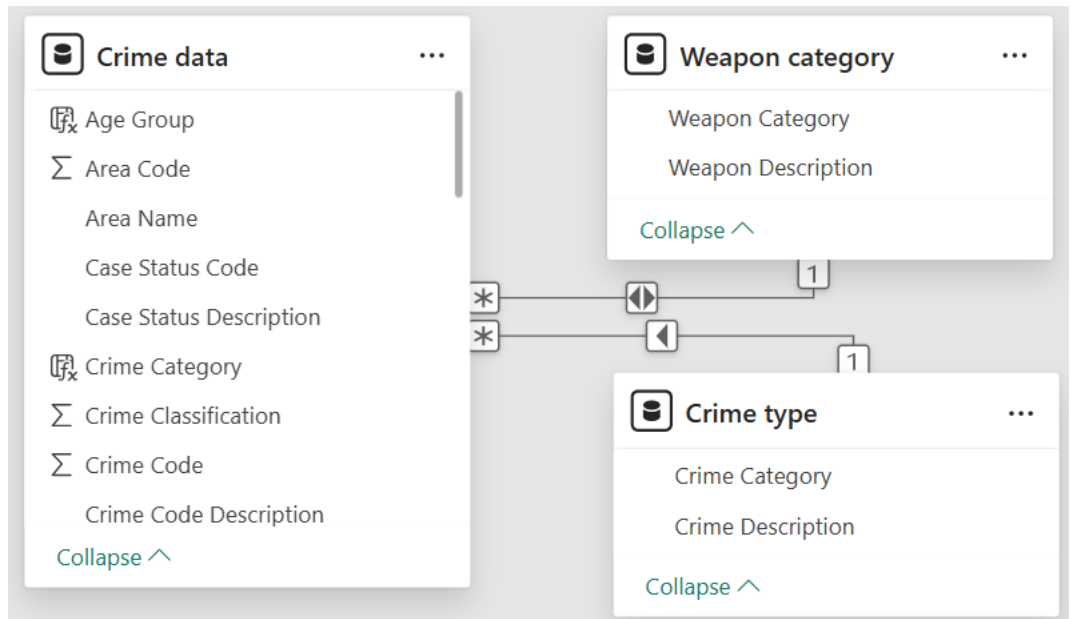
6. Data Pre-Processing (Excel / Power Query)

- **Tasks Performed**
 - The dataset was preprocessed in **Microsoft Excel** to improve data quality and prepare it for analysis in Power BI.
 - Renamed unclear column names with meaningful names.
 - Converted **Date Reported** and **Date Occurred** into Date format.
 - Converted **Time Occurred** from numeric values to standard Time format.
 - Standardized **Victim Gender** values (M → Male, F → Female, Others → Unknown).
 - Categorized detailed weapon descriptions into simplified weapon categories.
 - Assigned appropriate data types to all columns.
 - Validated and cleaned the dataset for consistency.
 - Imported the cleaned dataset into Power BI for further analysis.
 - Created lookup tables (Dimension Tables) for Crime Type and Weapon Category.
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7. Data Modelling and DAX (Power BI)

- **Data Model**
- A **Star Schema** data model was created by connecting the main Crime Data table with lookup (Dimension) tables.
- **Relationships Created:**
- | ○ Main Table | ○ Lookup Table | ○ Relationship |
|---------------------|-------------------------|-----------------------|
| ○ Crime Data | ○ Crime Type Table | ○ Crime Code |
| ○ Crime Data | ○ Weapon Category Table | ○ Weapon Used Code |
- Relationship Type: **One-to-Many**
- Cross Filter Direction: **Single**

Table Relationships



- **Calculated Columns**

- Crime Month
- Crime Month Order
- Crime Month Year

- **DAX Measures**

- **DAX Measure**

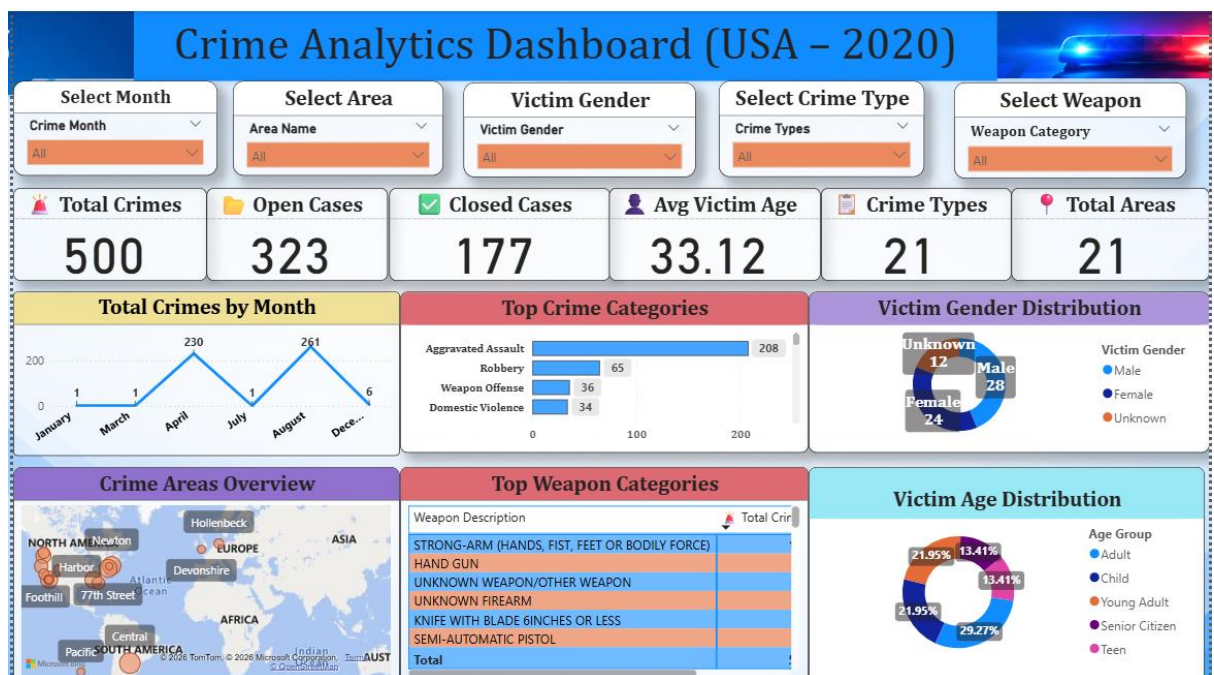
- Total Crimes
 - Open Cases
 - Closed Cases
 - Average Victim Age
 - Crime Types
 - Total Areas

- **Purpose**

- Counts total crime records
 - Counts unresolved cases
 - Counts resolved cases
 - Calculates average victim age
 - Counts unique crime categories
 - Counts reporting areas
 - These DAX measures were used to create KPI cards and support interactive dashboard analysis.
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8. Analysis and Visualizations (Power BI)

- **Dashboard Features**
- The Power BI dashboard was designed using interactive visuals to analyze crime data effectively.
- **Visualizations Used:**
- KPI Cards (Total Crimes, Open Cases, Closed Cases, Average Victim Age, Crime Types, Total Areas)
- Line Chart – Monthly Crime Trend
- Horizontal Bar Chart – Top Crime Categories
- Donut Chart – Victim Gender Distribution
- Donut Chart – Victim Age Distribution
- Map – Crime Areas Overview
- Table – Top Weapon Categories
- Interactive Slicers (Crime Month, Area Name, Crime Type, Victim Gender, Weapon Category)
- **Dashboard Features**
- Interactive Filters and Slicers
- Cross Filtering and Cross Highlighting
- Clear Titles and Data Labels
- User-Friendly Dashboard Layout
- Consolidated Report Page



9. Insights & Conclusions

Key Findings

○ 1. Overall Crime Statistics

- Total Crime Records: **500**
 - Open Cases: **323 (64.60%)**
 - Closed Cases: **177 (35.40%)**
 - Average Victim Age: **33.12 Years**
 - Crime Categories: **21**
 - Reporting Areas: **21**
 - **Insight:** Most reported cases remain open, indicating the need to improve investigation and case resolution processes.
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2. Monthly Crime Trend

- **August** recorded the highest number of crimes (**261**).
 - **April** recorded the second-highest number of crimes (**230**).
 - **Insight:** Crime incidents were higher during certain months, indicating seasonal variations in crime patterns.
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3. Top Crime Categories

- Aggravated Assault – **208**
 - Robbery – **65**
 - Weapon Offense – **36**
 - Domestic Violence – **34**
 - **Insight:** Aggravated Assault was the most frequently reported crime category, followed by Robbery.
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4. Victim Gender Distribution

- Male – **28%**
 - Female – **24%**
 - Unknown – **12%**
 - **Insight:** Male victims accounted for the highest percentage of identified victims.
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5. Investigation Status

- Open Cases – **323**
 - Closed Cases – **177**
 - **Insight:** More than half of the reported cases are still under investigation, highlighting the need to improve case closure rates.
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6. Top Weapon Categories

- Strong-arm Methods – **106**
 - Handgun – **74**
 - Unknown Weapon – **36**
 - Unknown Firearm – **35**
 - **Insight:** Strong-arm methods and handguns were the most commonly used weapons in reported crimes.
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Descriptive Analysis

- Analyzed **500** crime records across **21** reporting areas.
- Identified crime trends, victim demographics, weapon usage, and investigation status.

Diagnostic Analysis

- Aggravated Assault was the most common crime.
- August recorded the highest crime incidents.
- Most reported cases remain open.

Predictive Analysis

- High-crime areas may continue to experience increased crime if preventive measures are not implemented.
- Historical crime trends can help predict future crime patterns.

Prescriptive Analysis

- Increase police patrols in high-crime areas.
 - Improve investigation efficiency to increase case closure rates.
 - Strengthen public safety awareness programs.
 - Use dashboard insights for better resource allocation and decision-making.
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10. Conclusion

- The **Crime Analytics Dashboard** successfully demonstrates the complete data analytics process using **Microsoft Excel** and **Microsoft Power BI**. The project involved data cleaning, transformation, data modelling, DAX calculations, and interactive dashboard development to analyze crime data effectively.
- The dashboard provides valuable insights into crime trends, victim demographics, weapon usage, and investigation status, helping users make informed, data-driven decisions. This project highlights the effectiveness of Excel and Power BI in transforming raw data into meaningful visual reports for public safety and business intelligence.